COURSEWORK AI-USE DOCUMENTATION

FitLife 360
ChatGPT Working Session Log

October 25, 2026 | Q1-Q5 analysis, troubleshooting, interpretation, and submission review

Session boundary. This log begins with the first Q1 DiD output. Earlier assignment-planning discussion is excluded.

Record format. This is a chronological, organized chat log, not a verbatim export. Concise user prompts and numerical results are reproduced; long pasted code and screenshots are excerpted or described; assistant guidance is condensed.

Documented model results

Question / model	Estimated daily effect	95% interval
Q1 - Difference-in-differences	+5.381	[1.834, 8.927] confidence
Q2 - CausalImpact, Denver only	+8.0	[7, 9] credible
Q3 - CausalImpact + 3 cities	+4.5	[3.1, 5.8] credible
Q4 - CausalImpact + cities + crypto	+5.0	[2.6, 6.8] credible

How ChatGPT contributed

ChatGPT identified webinar examples, explained statistical output, proposed and corrected R code, debugged data transformations, helped interpret model results, reviewed the student's writing, and evaluated the notebook against the rubric and prior feedback.

The student ran code in RStudio, supplied model outputs and plots, selected the preferred model and rollout recommendation, and wrote initial interpretation paragraphs. AI-generated code examples and wording corrections are documented below.

Q1 | DiD regression and confidence interval

User

okay on q1. Here is what I have for q1
Call:
lm(formula = downloads ~ treated * post, data = df)
Coefficients:
(Intercept) 33.405; treated -14.548; post 2.429; treated:post 5.381

Assistant (condensed)

Identified treated:post as the DiD treatment estimate (+5.381 downloads/day), explained other coefficients, and suggested calculating confint(did, 'treated:post', level = 0.95).

User

> did <- lm(downloads ~ treated * post, data = df)
> confint(did, 'treated:post', level = 0.95)
treated:post 1.834472 8.927433

Assistant (condensed)

Confirmed the conventional OLS 95% confidence interval [1.834, 8.927], noted it excludes zero and cautioned about repeated observations within cities.

Q1 | Plot construction and parallel trends

User

what code lines from the webinar is the plot?

Assistant (condensed)

Directed the user to webinar4.qmd Q1 Benchmarks (approximately lines 80-92) for group_by(date, treated), mean(downloads) and ggplot; also referenced later time-series examples.

User

[Uploaded an initial average-downloads plot and R output; graph displayed Denver below controls and growing more rapidly. R displayed a harmless summarise() grouping message.]

Assistant (condensed)

Asked for a vertical line at treatment onset; suggested geom_vline(xintercept = min(df$date[df$post == 1]), linetype = 'dashed').

User

I am confused. what should my whole code block look like

Assistant (condensed)

Provided a complete Q1 block with did <- lm(...), confint(...), grouped means, geom_line(), geom_vline(), and axis labels.

User

[Uploaded a corrected plot with dashed July 15 treatment line and the updated R output.]

Assistant (condensed)

Interpreted Denver's faster pre-treatment growth relative to control cities; warned the parallel trends assumption looked questionable.

User

but denver overall has less downloads?

Assistant (condensed)

Clarified the distinction between level and trend: Denver had fewer downloads overall, but its increase was faster. Explained that DiD compares changes, not initial levels. Used the difference of increases (about 7.81 - 2.43 = 5.38).

User

okay so what do I need to answer for q1

Assistant (condensed)

Summarized the three Q1 requirements: plot; DiD estimate with 95% CI; explain pre-treatment trend divergence and risk of bias.

User

Denver had an estimated treatment effect of 5.381 additional downloads daily. The confidence interval excludes zero, which indicates a significant statistical positive estimate. Before treatment began, Denver had less downloads overall but the number of downloads was already increasing faster than average. Because of this, there are concerns about parallel trends assumption because controls and Denver did not have similar pretreatment trending.

Assistant (condensed)

Suggested explicitly including [1.834, 8.927], saying 'faster than control cities,' and adding that DiD may overstate the effect.

User

DiD ATE: 5.381, 95% Confidence Interval: [1.834, 8.927].

Denver had an estimated treatment effect of 5.381 additional downloads daily. The confidence interval excludes zero, which indicates a significant statistical positive estimate. Before treatment began, Denver had less downloads overall but the number of downloads was already increasing faster than the controls. Because of this, there are concerns about parallel trends assumption because controls and Denver did not have similar pretreatment trending and the DiD may overstate the treatment's true effect..

Assistant (condensed)

Approved the interpretation with minor grammar recommendations and repeated the standard-OLS uncertainty caveat.

Q2 | Univariate CausalImpact

User

q2 where to find the code I need in webinar

Assistant (condensed)

Located webinar4.qmd ## CausalImpact, approximately lines 339-376: select treatment response, create zoo series, define intervention dates and fit model.

User

response <- filter(df, product == 'Denver')$Y)
(dates <- filter(df, product == 'Denver')$date)
(data <- zoo(response, dates))
(pre <- as.Date(c('2021-07-01', '2021-07-14')))
(post <- as.Date(c('2021-07-15', '2021-08-01')))
impact <- CausalImpact(data, pre, post)
summary(impact)
plot(impact)
syntax errors

Assistant (condensed)

Identified extra parenthesis and incorrect webinar variable names; replaced product and Y with treated/downloads. Proposed denver <- df |> filter(treated == 1) |> arrange(date), response <- denver$downloads, dates <- denver$date.

User

[Executed revised code and uploaded CausalImpact three-panel plot. Posterior results: actual total 480; predicted 337; absolute average effect 8 [7, 9]; cumulative 143 [126, 162]; relative +43%; posterior tail-area p 0.001.]

Assistant (condensed)

Explained original, pointwise and cumulative plot panels; warned no-covariate counterfactual was nearly flat despite pre-treatment growth.

User

The estimated treatment effect of the Causal Impact without covariate series is 8 additional downloads per day. The 95% Confidence Interval is [7,9]

I think this is sufficient for q2 answer?

Assistant (condensed)

Confirmed Q2's short answer was sufficient provided the plot was included; corrected 'Confidence Interval' to Bayesian 'Credible Interval'.

Q3 | Add three control-city time series

User

q3 ARGhjjfhgipsnofgklnisdgniportkslundlvrkvg

Assistant (condensed)

Directed user to webinar approximately lines 382-417, which demonstrate reshaping panel data and fitting a multivariate CausalImpact model.

User

names(df)
unique(df$city)
[1] 'date' 'city' 'treated' 'downloads' 'post'
[1] 'Denver' 'San Francisco' 'Seattle' 'Los Angeles'

Assistant (condensed)

Adapted pivot_wider so dates are rows and cities are columns, with Denver as the first outcome series.

User

```
# Q3: Reshape data for CausalImpact with covariates

df_wide <- df |>
select(date, city, downloads) |>
pivot_wider(names_from = city, values_from = downloads) |>
arrange(date)
head(df_wide)
```

Assistant (condensed)

Requested the printed data preview; clarified that executing the code in the script does not necessarily display a result in the Console.

User

nothing is happening

Assistant (condensed)

Suggested print(head(df_wide)); explained Console > versus + prompt and possible reasons output might not appear.

User

[Reported head(df_wide): six daily rows, with columns date, Denver, San Francisco, Seattle, Los Angeles. July 1 values: 16, 27, 40, 30.]

Assistant (condensed)

Provided data_cov <- zoo(select(df_wide, -date), df_wide$date); impact_cov <- CausalImpact(data_cov, pre, post); summary(impact_cov); plot(impact_cov).

User

[Uploaded Q3 three-panel plot. Posterior results: average absolute effect +4.5 downloads/day, 95% credible interval [3.1, 5.8], cumulative +80.4 [55.5, 104.4], relative +20%.]

Assistant (condensed)

Compared Q2 +8.0 with Q3 +4.5. Noted that control-city predictors tracked Denver's historical downloads more closely and made the counterfactual more plausible, not guaranteed unbiased.

User

The estimated treatment effect of the Causal Impact with covariate series is 4.5 additional downloads per day. The 95% Confidence Interval is [3.1,5.8]

Assistant (condensed)

Approved with edits: use 'credible interval' and identify San Francisco, Seattle and Los Angeles as the covariates.

Q4 | Historical crypto data and expanded model

User

q4 njknjfgbdgipasmfgnsjdfbgiaorpjodgnesjdbfcgjkcnsldgjniog cjsd

Assistant (condensed)

Initially proposed external weather data as three daily covariates.

User

nlets do one of the examples

Assistant (condensed)

Switched to the assignment's cryptocurrency example using crypto2::crypto_history().

User

install cryto

Assistant (condensed)

Suggested install.packages('crypto2') and library(crypto2).

User

[Confirmed crypto2 installation and Bitcoin history retrieval. Data columns included timestamp, open, high, low, close, volume and market_cap.]

Assistant (condensed)

Proposed deriving three covariate series from btc: close, volume, market_cap; also warned that Bitcoin predictors may not be directly related to fitness-app downloads.

User

[Reported btc_cov head(): daily rows starting 2021-07-01 with date, close, volume, market_cap.]

Assistant (condensed)

Asked user to verify complete daily coverage and no missing values.

User

[1] 32
[1] 0

Assistant (condensed)

Provided join-and-fit steps: df_extended <- left_join(df_wide, btc_cov, by='date'); data_extended <- zoo(...); impact_extended <- CausalImpact(...). Warned highly related crypto series and short pre-treatment period could reduce robustness.

User

[Uploaded Q4 three-panel plot. Posterior results: average absolute effect +5.0, 95% credible interval [2.6, 6.8]; cumulative +90 [46.7, 123.0]; relative +24%.]

Assistant (condensed)

Compared all four model estimates, noting Q4's interval was wider than Q3's, and advised acknowledging possible weak predictor relevance.

User

The estimated treatment effect of the Causal Impact with covariate series + cryto is 5 additional downloads per day. The 95% Credible Interval is [2.6, 6.8]

Q5

Assistant (condensed)

Approved Q4 result with a typo correction and began Q5 guided writing.

Q5 | Guided explanation and recommendation

User

FitLife choose Denver as it's only treatment city to say on experiment expense. By using models that control for variables unique to time and city characteristics, we are able to produce a meaningful experiment that can replicate to other cities.

Assistant (condensed)

Explained financial-risk motive and cautioned a one-city result does not establish reproducibility across other cities.

User

The DiD estimate was less reliable due to concerns about parallel trend assumptions for Denver. Before treatment began, Denver was growing in daily downloads at a higher rate than the control cities. Becaues of this, DiD may overstate the treatment's true effect.

Assistant (condensed)

Validated parallel-trends reasoning and recommended noting the difficulty of averaging out city-specific effects when N = 1.

User

Causal Impact uses historical download trends to forecast what might have happened with advertising. Then compares treatment to the forecast to estimate treatment effect. We subsequently added additional variables like control cities' downloads and crypto factors to help with the forecasting and comparison to help improve identifying the true impact of the treatment.

Assistant (condensed)

Corrected 'with advertising' to 'without advertising'; explained counterfactual estimation and cautioned more covariates do not necessarily improve accuracy.

User

CausalImpact with control cities +4.5 downloads/day, 95% credible interval [3.1, 5.8], because the control-city downloads have a clearer relationship to Denver's downloads than cryptocurrency prices do.

Assistant (condensed)

Endorsed Q3 as the more defensible estimate and suggested that control cities help account for common download trends.

User

Move forward with pilot testing in multple cities to evaluate reporduciability in other cities. after 30 days, if results are positive, move to full rollout.

Assistant (condensed)

Recommended caution about automatic rollout: pilot results should be statistically convincing and financially justify advertising costs.

User

please put my word for word with minor gramatical corrections and concept corrections in one output

Assistant (condensed)

Compiled a complete Q5 response largely preserving the student's language, with grammar and factual corrections.

Notebook and rubric review | Final submission checks

User

[Pasted full FitLife 360 Quarto notebook and asked:] evaluate for core gaps against the rubric. consider last week's feedback to identify any area that might be exposed for similar feedback

Assistant (condensed)

Identified Q3's missing df_wide creation and Q4's missing btc_cov creation; suggested targeted code additions. Compared gaps to last week's feedback about confounding, city effects, and code annotations; recommended a clean render, short explanations and an AI-use disclosure.

User

[Pasted revised notebook including df_wide and btc_cov creation. Noted:] I haven't completed the ai disclosure yet

Assistant (condensed)

Confirmed missing object definitions were fixed; recommended removing instructional template text, adding comments, checking YAML and URL syntax, rendering from a clean R session, and transparently disclosing code help, debugging, writing feedback, and specific prompts.

Appendix | Representative assistant-provided R code

Substantive implementation assistance provided during the session; these excerpts are not a complete copy of the student's final notebook.

Q1: DiD estimate, confidence interval and intervention plot

```
did <- lm(downloads ~ treated * post, data = df)
confint(did, 'treated:post', level = 0.95)
df |>
group_by(date, treated) |>
summarize(avg_downloads = mean(downloads), .groups = 'drop') |>
mutate(treated = factor(treated)) |>
ggplot(aes(date, avg_downloads, col = treated)) +
geom_line() +
geom_vline(xintercept = min(df$date[df$post == 1]),
linetype = 'dashed') +
theme_minimal()
```

Q2: Univariate CausalImpact

```
denver <- df |> filter(treated == 1) |> arrange(date)
response <- denver$downloads
dates <- denver$date
data <- zoo(response, dates)
pre <- as.Date(c('2021-07-01', '2021-07-14'))
post <- as.Date(c('2021-07-15', '2021-08-01'))
impact <- CausalImpact(data, pre, post)
summary(impact)
plot(impact)
```

Q3: Reshaping and model with three control cities

```
df_wide <- df |>
select(date, city, downloads) |>
pivot_wider(names_from = city, values_from = downloads) |>
arrange(date)
data_cov <- zoo(select(df_wide, -date), df_wide$date)
impact_cov <- CausalImpact(data_cov, pre, post)
summary(impact_cov)
plot(impact_cov)
```

Q4: Bitcoin predictors and expanded model

```
btc <- crypto_history(limit = 1,
start_date = '2021-07-01', end_date = '2021-08-01')
btc_cov <- btc |>
mutate(date = as.Date(timestamp)) |>
select(date, close, volume, market_cap) |>
arrange(date)
df_extended <- df_wide |> left_join(btc_cov, by = 'date')
data_extended <- zoo(select(df_extended, -date),
df_extended$date)
impact_extended <- CausalImpact(data_extended, pre, post)
summary(impact_extended)
plot(impact_extended)
```

Disclosure note

This record documents substantive ChatGPT help with code adaptation, debugging, statistical interpretation, writing revision and rubric review. It should accompany an accurate disclosure in the submitted Quarto notebook. It is a structured session log, not a certified verbatim transcript.